State Aggregation in Dynamic Graph Recovery: Separating Structural Ranking from Temporal Discrimination

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Abstract

Post-estimation aggregation of dynamic graph estimates can improve edge ranking while weakening recovery of when edges are active. We study this distinction through paired interventions that hold local lagged-edge scores fixed while varying their temporal emission. Evaluation separates pooled AUROC, within-edge temporal AUROC and boundary detection, supported by exact diagnostic identities for static scores and two-state hard aggregation. Across 100 independent paired simulations, hard aggregation increases pooled AUROC from 0.727 to 0.780 but reduces within-edge temporal AUROC from 0.707 to 0.639 at the shorter-duration configuration. At the longer-duration configuration, its temporal gain is 0.038. Schedule, generator and window comparisons reveal configuration dependence, including changes in switch coverage. Shrinkage toward edgewise means improves pooled ranking without changing temporal order. Retaining window-level residuals restores comparisons lost through hard-output ties, with a measurable rank jump at a near-zero positive residual weight. Boundary assessments also depend on zero-score conventions and threshold references. A segmentation-and-refit comparator improves pooled ranking without an established within-edge temporal gain. Together, these findings in two-state linear systems motivate separate evaluation of structural ranking, temporal discrimination and boundary detection in dynamic graph recovery.

Keywords: Dynamic graph recovery; temporal discrimination; boundary detection; state aggregation; evaluation; switching autoregression.

1 Introduction

A time-varying graph estimator must recover which relationships exist and when they are active. An aggregation step can improve the first objective while weakening the second. This creates a practical ambiguity: a better ranking of edges may reflect more accurate long-run structure even when the output loses information about changes. Evaluation must resolve that ambiguity before a structural improvement can guide interpretation of a dynamic system.

Grouping local connectivity estimates into recurring states is an established strategy. Sliding-window functional-connectivity studies use representative states to describe recurrence, reproducibility and task-related variation [Allen et al., 2014, Abrol et al., 2017, Chang et al., 2026]. For graph recovery, the corresponding question is how replacing local estimates by group summaries changes the recovery of individual edges across time. Shared observations support a controlled comparison: hold the local scores fixed and vary grouping, score emission and the decision to execute aggregation.

We distinguish three evaluation targets. Pooled AUROC compares positive and negative edge–window entries, mixing comparisons within and across edges. Within-edge temporal AUROC compares an edge's active and inactive windows. Boundary detection evaluates score changes between adjacent windows. A static occupancy example and finite-score identities connect these targets to the information that different emission rules preserve.

The distinctions change the empirical assessment of aggregation. Across 100 independent paired simulations at a short-duration configuration, hard aggregation raises pooled AUROC from 0.727 to 0.780 while reducing within-edge temporal AUROC from 0.707 to 0.639. At a longer-duration configuration, the within-edge gain is positive. Shrinkage toward each edge's temporal mean also improves pooled ranking while preserving its temporal order, showing that this structural benefit can arise without state identification. Residual controls then locate the comparisons lost when hard

aggregation assigns one score to every window in a group.

Contributions. Three contributions organise the study.

- An evaluation protocol separates structural ranking, within-edge temporal discrimination and boundary detection on shared inputs and local scores. Static, raw, size-matched and oracle controls reveal when an apparent improvement depends on the evaluation target.
- Paired simulations identify contrasting temporal effects of hard aggregation at two duration configurations. Schedule, generator and window comparisons show how these effects depend on the evaluated configuration; score-preserving controls identify structural gains that occur without grouping.
- Diagnostic identities connect hard-emission performance to state-assignment discrimination and graph contrast, and decompose residual gains into same-group and cross-group comparisons. Boundary sensitivities and a segmentation/refit comparison show how score conventions affect the resulting assessment.

The experiments use two-state linear switching systems with local Lasso and OLS scores. This controlled setting links the evaluation analysis to paired changes in graph recovery while exposing the roles of window resolution, score emission and gate acceptance.

2 Related work

Post-estimation grouping is exemplified by dynamic functional-connectivity analysis. Allen et al. [2014] cluster regularised windowed covariance matrices into recurring states, using Manhattan distances and cluster-median maps. Abrol et al. [2017] examine replication of state profiles and temporal summaries, including dwell times and transitions, with hard clustering and fuzzy meta-

states. Task-fMRI work also relates clustered connectivity states to ongoing events and behaviour [Chang et al., 2026]. These are meaningful validation targets for functional association. They differ from recovery of the temporal support of individual directed lagged edges.

Using simulated networks with known transitions, Shakil et al. [2016] already evaluate correlation fidelity, state identification and transition recovery as window parameters and state durations change. Our comparison isolates the paired gain of aggregating shared local scores over raw windows on within-edge temporal AUROC. It distinguishes state-level temporal summaries from within-edge score ordering, using score-preserving and residual controls to attribute a change in assessment. The distinction is the effect of aggregation on shared local scores, evaluated separately from state-level summaries and transition detection.

Lagged structure discovery includes stationary Granger-style models [Granger, 1969, Runge et al., 2019]. Joint regime and graph estimators include Regime-PCMCI, CASTOR and SPACE-TIME [Saggioro et al., 2020, Rahmani and Frossard, 2025, Mameche et al., 2025]. Jointly fitting state assignments and graph models changes the local estimates as well as their grouping. Aggregating fixed scores is a separate intervention. Penalised changepoint optimisation [Killick et al., 2012] and VAR break models [Bai and Perron, 1998] motivate the segmentation comparisons in Supplement Section S6.

Multidimensional evaluation can expose differences that a single structural score conceals. Velev and Lessmann [2026] evaluate causal discovery from observational i.i.d. data through structural recovery and the suitability of estimated graphs for causal inference under varying assumptions. Here the evaluation object is a time-varying graph: rankings across edges and temporal discrimination within changing edges need separate accounting. The classical bipartite-ranking principle [Clémençon et al., 2008] underlies our static occupancy ceiling; its role here is a task-specific diagnostic.

Evaluation choices also alter comparisons in dynamic link prediction [Poursafaei et al., 2022], where simple memory baselines and the choice of negative examples matter. That task observes a graph and predicts future links. We instead infer latent lagged support from multivariate series and ask what grouping windows preserves. Constant-score controls and within-edge temporal AUROC make this difference assessable.

3 Problem setup and controlled aggregation framework

Let $X_{1:T} \in \mathbb{R}^{T\times N}$ follow an unobserved state sequence z_t. The binary tensor $A_{k,\ell,i,j}$ marks the lag-ℓ support from source i to target j in state k. Regression includes self- and cross-lags, but edge evaluation and clustering exclude self-lag scores. Union evaluation uses $A_{\cup,\ell,i,j} = \mathbf{1}\{\exists k : A_{k,\ell,i,j} = 1\}$; regime evaluation matches state identities before scoring; time-indexed evaluation compares each emitted graph with the true state at its window centre. AUROC, AUPRC and precision inherit the meaning of the chosen target and indexing.

3.1 A shared-score aggregation experiment

We investigate post-estimation state aggregation using a controlled window-based framework. Local lagged-edge scores are held fixed while grouping, score emission and the acceptance rule are varied to isolate their effects on graph recovery. Window length h and stride s give $M = 1 + \lfloor (T-h)/s \rfloor$ windows. Each window is standardised per variable; one Lasso per target is fitted to standardised lagged inputs with $\alpha = 0.005$. Absolute coefficients form scores $S_{w,e}$. Self-lags enter regression but are excluded from the evaluated edges and clustering features. Non-self scores are standardised across windows and clustered into a supplied $K = 2$ groups by K-means. The auxiliary OLS scorer uses the same windows, preprocessing and evaluation. Supplement Section S2 gives the fitting procedures; Supplement Section S4 reports numerical-failure rules and coverage.

For candidate labels $\widehat{z}_w$, hard aggregation emits the mean score of each window's group. We distinguish three execution rules. *Forced hard aggregation* always emits that candidate. *Gated hard aggregation with static fallback* emits it when a fixed inertia/coherence gate accepts and otherwise emits the same window-maximum graph at every window. *Gated hard aggregation with raw fallback* uses the same acceptance decision and emits the raw window scores on rejection. The gate evaluates relative inertia improvement, adjacency agreement adjusted for group proportions, and minimum group size; its full definition is in Supplement Section S2. These rules separate candidate quality from the effects of acceptance and rejection output.

Raw windows emit S_w directly. Static mean emits one temporal-mean graph. Oracle-label aggregation averages by the true window-centre state without refitting. In Study A and the earlier selected seven-window comparison at $D = 600$, size-matched random grouping enumerates every unlabelled bipartition with the candidate's group sizes and averages its metrics within each simulation repeat. These controls share observations and local scores; the oracle additionally supplies true state assignments.

3.2 Generators and paired interventions

The simulations use $N = 8$ variables, maximum lag $L = 3$ and independent Gaussian innovations with standard deviation 0.10. Studies A and B use an asymmetric support construction. State A has non-self support probability 0.18 and no self-lags. State B retains each A edge with probability 0.5 and can add edges, including self-lags, outside A's support; its non-self marginal probability is 0.1638 before empty-support safeguards. Coefficients are redrawn between states, including on shared support. These properties are fixed throughout A and B. Study C jointly changes support symmetry, self-lags, shared coefficients and the stabilisation rule to assess a separate generator construction. Supplement Section S2 specifies the A/B mechanism; Section S4.2 specifies the C

mechanism and its window configurations.

Each independent repeat redraws the graph mechanism, initial history and innovations. Conditions within a repeat share all three, making their differences paired. After 100 burn-in samples, records alternate between two states under fixed-duration, random-phase or irregular-duration schedules. At fixed series length, changing duration also changes the number and coverage of observed transitions. Window length, stride, window count and realised switch counts are therefore reported alongside duration. Post-scaling companion radii are 0.381–0.676 in A and 0.404–0.627 in B, and all 500 observed A/B trajectories are finite. Supplement Section S8 reports the associated trajectory diagnostics.

OpenAI's ChatGPT and Codex assisted programming and data analysis. Their use for English translation and LaTeX formatting corrections is disclosed in the end-matter declaration. Figures are generated by reproducible Python scripts from the numerical outputs and the specified finite example. The reproducibility archive supplies these scripts and their inputs. The author reviewed and verified the AI-assisted outputs as appropriate.

4 Evaluation and mechanism analysis

4.1 Pooled ranking and within-edge temporal discrimination

For edge occupancy $\pi_e = M^{-1}\sum_{w=1}^{M} A_{w,e}$ and constant scores s_e, writing $H(u) = \mathbf{1}(u > 0) + \frac{1}{2}\mathbf{1}(u = 0)$, pooled AUROC is

$$\mathrm{AUC}_{\mathrm{static}} = \frac{\sum_{e,f} \pi_e(1-\pi_f)H(s_e - s_f)}{PQ}, \qquad P = \sum_e \pi_e, \quad Q = \sum_e (1-\pi_e). \tag{1}$$

It depends on frequencies rather than activation times. Sorting edges by π_e is optimal among constant scores: an adjacent exchange changes the numerator by the difference in occupancies.

Classical optimal ranking thus gives

$$\mathrm{AUC}^{*}_{\mathrm{static}} = \tfrac{1}{2} + \frac{\sum_{e,f} |\pi_e - \pi_f|}{4PQ}. \tag{2}$$

This ceiling characterises the ranking attainable from occupancy alone.

For general time-indexed scores, the positive–negative comparison pairs split exactly into same-edge and cross-edge parts:

$$\mathrm{AUC}_{\mathrm{pooled}} = \omega \mathrm{AUC}_{\mathrm{same}} + (1-\omega)\mathrm{AUC}_{\mathrm{cross}}, \qquad \omega = \frac{\sum_e n_e^+ n_e^-}{(\sum_e n_e^+)(\sum_e n_e^-)}. \tag{3}$$

Here $n_e^+ = \sum_{w=1}^{M} A_{w,e}$ and $n_e^- = M - n_e^+$ count the positive and negative window labels of edge e on the same evaluation grid. Cross-edge comparisons can also depend on time. Equation (3) therefore identifies the mixture of comparisons in pooled AUROC, with its same-edge weight determined by the evaluation labels.

Let $\mathcal{E}_\Delta$ contain non-self lagged edges whose binary label changes in the evaluation windows. Within-edge temporal AUROC on changing edges is

$$\mathrm{T-AUC}_\Delta = |\mathcal{E}_\Delta|^{-1} \sum_{e \in \mathcal{E}_\Delta} \mathrm{AUROC}\big((A_{w,e})_w, (\widehat{S}_{w,e})_w\big). \tag{4}$$

An edge without both positive and negative evaluation labels is not scored; undefined targets remain missing. Constant emission scores 0.5 whenever this target is defined. Figure 1 illustrates why high pooled ranking need not imply within-edge temporal discrimination. Equation (4) measures the ordering of an edge's active and inactive windows. It does not measure distance to a switch boundary, and it excludes edges whose support is constant. Permuting scores within the active windows, or within the inactive windows, leaves it unchanged while potentially changing the locations of thresholded errors. We therefore also evaluate boundary detection and variation on stable edges.

Boundary-slot detection. A slot is the interval between consecutive window centres. It is positive for an edge if its true support changes at least once in that interval, and the predicted change strength is the absolute difference between consecutive emitted scores. This target evaluates change detection at stride resolution. We report boundary-detection AP and F1, together with false changes on stable edges. Standard AP includes the zero-score group; positive-change AP treats zero strength as abstention. Supplement Section S1 summarises slot targets; Section S5 defines threshold references and AP conventions, compared here in Section 5.6.

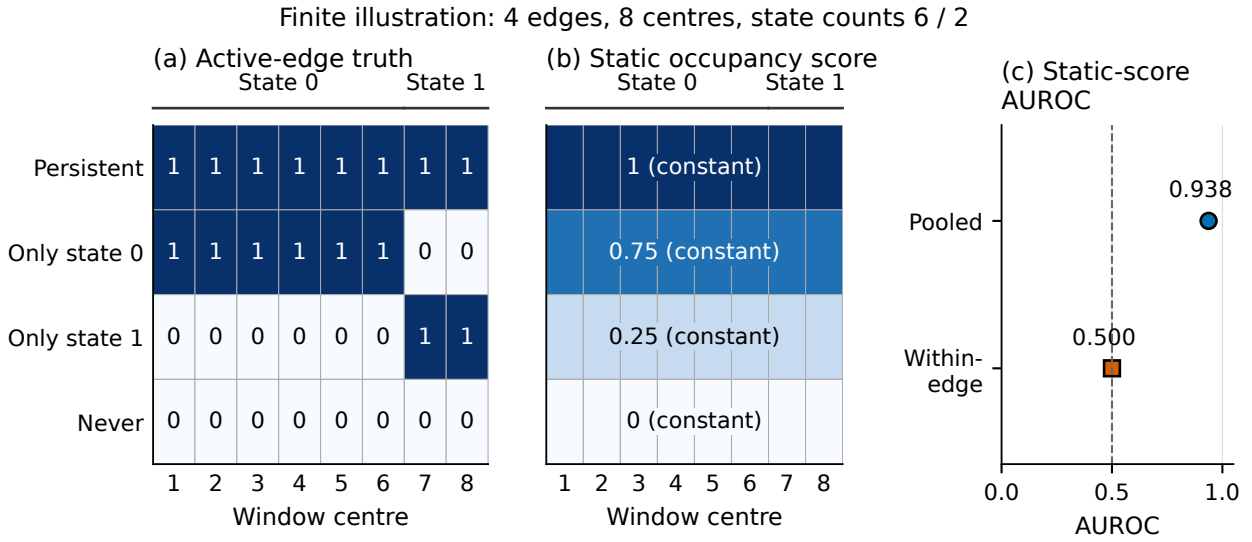


Figure 1: A deterministic evaluation example, not an empirical estimate. Time-indexed truth is compared with constant occupancy scores. The static predictor can rank frequent edges accurately while having within-edge temporal AUROC 0.5. The underlying truth and scores accompany the figure.

4.2 Two-state hard emission and relative gain

With two true states and two fixed predicted maps, align state numbering once. Define state-assignment discrimination $J = \Pr(\widehat{z} = 1 \mid z = 1) - \Pr(\widehat{z} = 1 \mid z = 0)$ and, for changing edges,

$$C_{\text{edge}} = |\mathcal{E}_\Delta|^{-1} \sum_{e \in \mathcal{E}_\Delta} \operatorname{sign}[(A_{1,e} - A_{0,e})(G_{1,e} - G_{0,e})] \,. \tag{5}$$

The two possible emitted scores per edge make its AUROC depend only on state-assignment discrimination and the direction of the map difference, yielding

$$\mathrm{T-AUC}_{\Delta} = \tfrac{1}{2} + \tfrac{1}{2} JC_{\mathrm{edge}}. \tag{6}$$

The identity is invariant to exchanging predicted state names. Across repeats, the mean of products is required; marginal means must not be multiplied. For the same repeat set, with T denoting Eq. (4),

$$\overline{\Delta}_{\mathrm{temporal}} = \overline{T}_{\mathrm{hard}} - \overline{T}_{\mathrm{raw}} = \tfrac{1}{2}\left[\overline{JC_{\mathrm{edge}}} - (2\overline{T}_{\mathrm{raw}} - 1)\right]. \tag{7}$$

The identity makes relative gain explicit: the mean grouping–contrast product must exceed the raw score's temporal-ranking reference for a positive mean hard-minus-raw difference. Both factors use ground-truth states and changing edges, so this is a retrospective diagnostic of the evaluated configuration. The occupancy and comparison identities apply to finite arrays independently of the local scorer; the factorisation requires two true states and two emitted maps. Supplement Section S3 gives the derivations and auxiliary window-mixing and risk analyses.

The specified gate has a separate geometric property. With seven windows, optimal two-group inertia improvement is at least 1/6, exceeding its 0.12 acceptance threshold even without state structure. The bound concerns the optimal partition; actual K-means fits and gate outcomes are measured separately. This motivates reporting forced candidates and both fallback rules alongside acceptance rates (Supplement Section S3).

4.3 Score-preserving controls and the fallback decision

For each edge, let $\overline{S}_e = M^{-1}\sum_w S_{w,e}$. Global shrinkage

$$S^{\mathrm{shrink}}_{w,e} = \lambda S_{w,e} + (1-\lambda)\overline{S}_e, \qquad 0 < \lambda \leq 1, \tag{8}$$

preserves every within-edge comparison, including ties, in exact arithmetic: $S^{\text{shrink}}_{u,e} - S^{\text{shrink}}_{v,e} = \lambda(S_{u,e} - S_{v,e})$. Hence it preserves Eq. (4) for any graph generator and any local scorer. Across different edges, the edge-specific offsets can change pooled rankings. This gives a non-clustering control for structural improvement with temporal order preserved: any pooled gain under this intervention arises through changed cross-edge comparisons.

We also retain window residuals around the candidate hard emission $H_{w,e}$:

$$S^{\text{residual}}_{w,e} = H_{w,e} + \lambda(S_{w,e} - H_{w,e}). \tag{9}$$

This interpolates between hard ($\lambda = 0$) and raw ($\lambda = 1$). Its edgewise ordering across candidate groups need not equal raw ordering. Within a common candidate group, $H_{u,e} = H_{v,e}$ and the residual score difference is $\lambda(S_{u,e} - S_{v,e})$. Thus every positive weight restores that group's raw ordering, whereas hard emission ties those comparisons. AUROC can jump at zero residual weight; it need not improve smoothly with the retained amplitude. We account for same-group and cross-group pair contributions separately, including a diagnostic near-zero weight. We report all fixed values $\lambda \in \{0.25, 0.5, 0.75, 1\}$ for both controls, alongside the hard endpoint, without choosing a value using the evaluation truth. The implementation checks actual finite-precision order and tie changes.

Let I indicate the fixed gate's acceptance and T the within-edge temporal AUROC of one repeat. Emitting raw on rejection gives

$$\Delta_{\text{raw fallback}} = I(T_{\text{hard}} - T_{\text{raw}}), \qquad \Delta_{\text{static fallback}} = I(T_{\text{hard}} - T_{\text{raw}}) + (1 - I)(\tfrac{1}{2} - T_{\text{raw}}). \tag{10}$$

Thus the difference between the two fallback rules isolates information lost by constant emission on rejection. The mean raw-fallback gain is the acceptance rate times the accepted subset's own mean hard-minus-raw gain. This separates candidate selection from information discarded by the rejection output.

5 Controlled empirical study

5.1 Experimental design and statistical inference

Table 1 summarises the three simulation studies. Study A compares two duration configurations in 100 independent paired repeats. Pilot experiments informed the selected configurations; the primary contrasts and multiplicity adjustments were specified before the corresponding confirmatory simulations. Its primary family contains four Lasso contrasts: gated hard minus raw at each duration; their within-repeat difference $\Gamma = \mathbb{E}[\Delta(600) - \Delta(300)]$; and hard minus size-matched random grouping at $D = 600$. Two-sided repeat-level t tests use Holm correction across these four contrasts. OLS and forced-hard comparisons are prespecified auxiliary analyses.

Study B uses 50 different independent repeats to compare two mean durations under fixed, random-phase and irregular schedules. Its six forced-minus-raw comparisons form a secondary Holm6 family, and its three paired duration differences form a separate secondary Holm3 family. Study C is a separately specified robustness study with 30 further independent repeats, two durations and five window/stride settings. Studies A–C are reported separately; they were not externally preregistered. Supplement Section S4 gives their complete designs, seed ranges, contrasts and numerical coverage.

Additional score-emission controls use the fitted predictions from these studies. The AP, threshold and near-zero-residual sensitivity settings were chosen after A–C results were available, and the fixed segmentation/refit comparison uses existing Study C cases. These analyses are exploratory and remain outside the primary confirmatory family. All intervals are pointwise 95% percentile bootstrap intervals based on 10,000 repeat resamples; paired differences are computed within repeat before summarisation. Repeats, rather than windows, edges or random partitions, are the statistical units. Fit failures and undefined targets are reported with their denominators.

Table 1: Simulation designs. Within a repeat, conditions share graph coefficients, support, initial history and innovations. Window length/stride are denoted h/s; Study C's five settings assess window resolution under a separate generator. Primary, secondary and exploratory analyses use the distinct inference rules described in the text.

Study: repeats / seeds	Paired design	T	Window grid: h/s; M	Statistical role
A: 100 independent repeats	Fixed phase; $D = 300, 600$			Confirmation: gated Lasso Holm4
50000–50099	Two cells per repeat	1200	300/150; 7	Forced and OLS auxiliary
B: 50 independent repeats	Fixed/random/irregular			Secondary robustness
60000–60049	$\bar{D} = 300, 600$; six cells	3600	300/150; 23	Forced–raw Holm6; duration Holm3
			200/100; 35	
			300/150; 23	
			400/200; 17	Separate robustness study
C: 30 independent repeats	Symmetric generator		300/75; 45	Pointwise 95% intervals
70000–70029	$D = 300, 600$; ten cells	3600	300/300; 12	No confirmatory family

5.2 Structural ranking gains versus temporal discrimination

Hard aggregation improves pooled ranking while having opposite temporal effects at the two duration configurations (Table 2). At $D = 300$, pooled AUROC rises from 0.727 to 0.780, while within-edge temporal AUROC falls from 0.707 to 0.639. The paired temporal difference is -0.069 $[-0.090, -0.048]$, compared with $+0.038$ $[0.023, 0.053]$ at $D = 600$. Their direct paired difference is $\Gamma = +0.107$ $[0.079, 0.134]$. Hard aggregation also exceeds size-matched random grouping by $+0.226$ $[0.210, 0.243]$ at $D = 600$. All four primary comparisons pass Holm4 correction; the largest adjusted p value is 2.93×10^{-6}.

All 400 Lasso/OLS fits succeed and all temporal targets are evaluable. The gate accepts all 100/100 fits at both durations, so gated and forced hard outputs coincide here. The observed records contain four versus two switches, and the seven window centres contain three versus one switches, at $D = 300$ versus 600. Thus Figure 2 compares two complete duration/coverage configurations.

The prespecified OLS check yields the same contrast: hard-minus-raw is -0.071 $[-0.092, -0.051]$ at $D = 300$ and $+0.032$ $[0.017, 0.046]$ at $D = 600$, with paired difference $+0.103$ $[0.076, 0.129]$. Supplement Section S4 reports all contrasts, scores and fit diagnostics.

Table 2: Study A: mean pooled and within-edge temporal AUROC on the same independent repeats. All methods share local Lasso scores within each fit. Intervals and the four primary comparisons are shown in Figure 2; coverage and failure counts are reported in Supplement Section S4. Paired differences are computed before rounding: the displayed marginal means 0.639 and 0.707 do not determine the displayed difference −0.069.

	$D = 300$		$D = 600$	
Method	Pooled AUROC	Within-edge AUROC	Pooled AUROC	Within-edge AUROC
Raw windows	0.727	0.707	0.772	0.767
Static mean	0.806	0.500	0.841	0.500
Gated hard	0.780	0.639	0.839	0.805
Forced hard	0.780	0.639	0.839	0.805
Matched random	0.779	0.559	0.817	0.579
Oracle labels	0.803	0.797	0.846	0.848

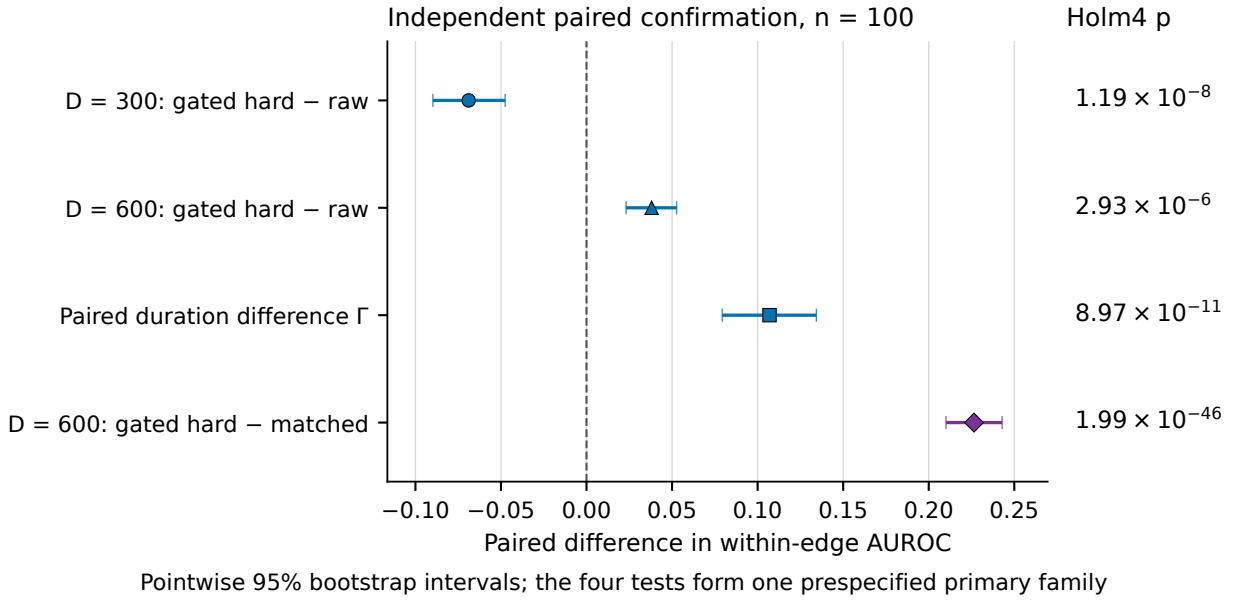


Figure 2: Study A: paired repeat differences and the four-comparison primary family. Intervals are pointwise 95% bootstrap intervals; p values are two-sided t tests with Holm4 adjustment. Duration differences are paired within the same mechanism/noise repeat.

5.3 Sensitivity to switching schedules and gate acceptance

Positive candidate effects at $D = 600$ persist across the three switching schedules (Figure 3). Forced hard-minus-raw within-edge temporal AUROC is +0.072 $[0.055, 0.089]$ for fixed phase, +0.072 $[0.050, 0.094]$ for random phase and +0.070 $[0.047, 0.091]$ for irregular runs; all three pass secondary Holm6 correction. At $D = 300$, the estimates are −0.025, +0.003 and −0.019, with intervals crossing zero. The paired duration differences are +0.097, +0.070 and +0.089, all passing the

separate secondary Holm3 family. All six cells have 50 successful, evaluable repeats (300 condition fits in total). The fixed-phase records contain eleven/five window-centre switches, extending the three/one-switch contrast of Study A.

Gate acceptance changes the effect of executing this candidate. Acceptance is 2/50, 0/50, 3/50 at $D = 300$ and 6/50, 6/50, 7/50 at $D = 600$ for fixed, random and irregular schedules. Gated hard aggregation with static fallback has negative gains throughout, from -0.214 to -0.198 (Figure 4). Section 5.4 examines how much of that loss comes from the rejection output.

The factor diagnostics in Table 3 account for the stronger forced candidates at the longer duration. Mean per-repeat JC_{edge} is approximately 0.666, 0.656 and 0.659 at $D = 600$, compared with 0.358, 0.400 and 0.363 at $D = 300$. State-assignment discrimination and graph-contrast concordance change together in these comparisons.

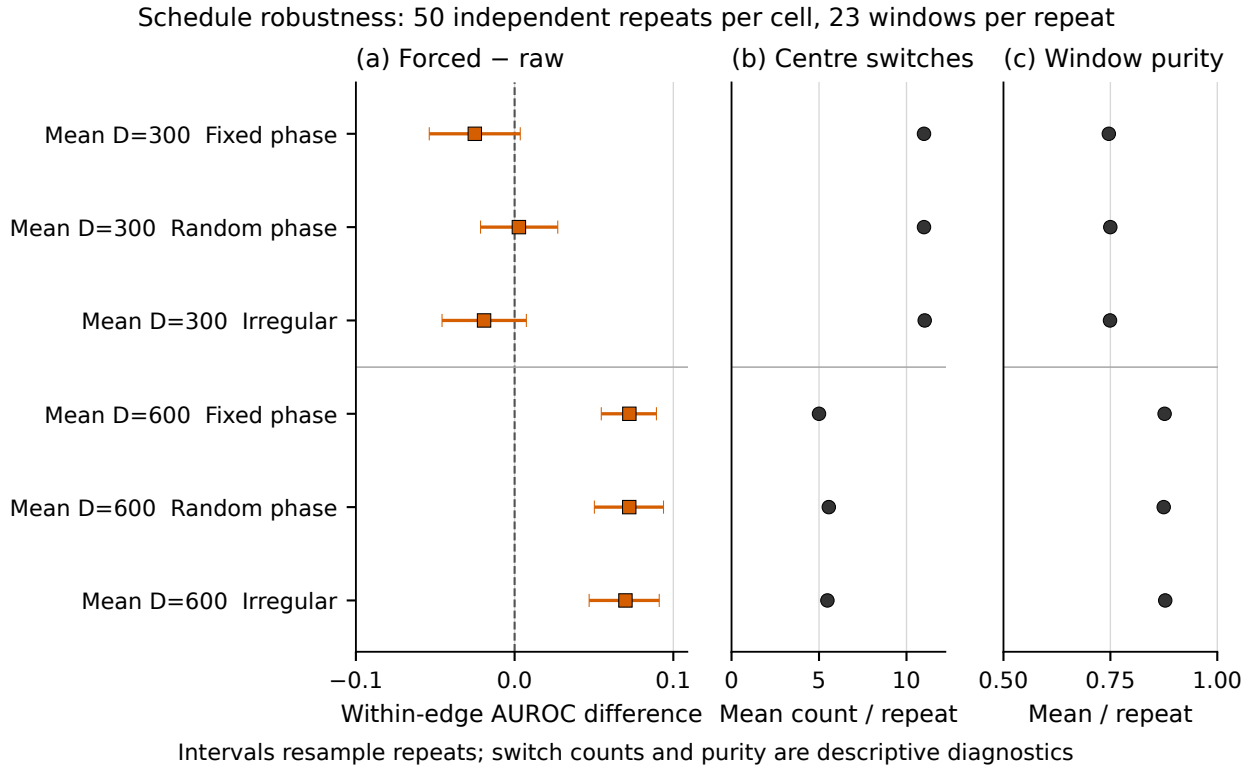


Figure 3: Study B: forced aggregation gains, realised window-centre switch counts and mean window purity for all six planned cells. Points/intervals use independent simulation repeats. The schedule comparisons are secondary robustness analyses.

Table 3: Candidate-hard factor diagnostics. Every row uses all evaluable forced candidates, including gate rejections. Per-repeat products are averaged across repeats; raw reference is $2\overline{T}_{\text{raw}} - 1$. Differences among jointly estimated factors are descriptive.

Study/cell	D or $\bar{D}$	n	$\overline{J}$	$\overline{C}_{\text{edge}}$	$\overline{JC_{\text{edge}}}$	$2\overline{T}_{\text{raw}} - 1$	Forced Δ
A: Fixed phase	300	100	0.550	0.406	0.277	0.415	-0.069
A: Fixed phase	600	100	0.888	0.677	0.610	0.534	+0.038
B: Fixed phase	300	50	0.525	0.592	0.358	0.408	-0.025
B: Random phase	300	50	0.545	0.687	0.400	0.395	+0.003
B: Irregular	300	50	0.510	0.651	0.363	0.402	-0.019
B: Fixed phase	600	50	0.817	0.812	0.666	0.521	+0.072
B: Random phase	600	50	0.807	0.807	0.656	0.511	+0.072
B: Irregular	600	50	0.786	0.828	0.659	0.520	+0.070

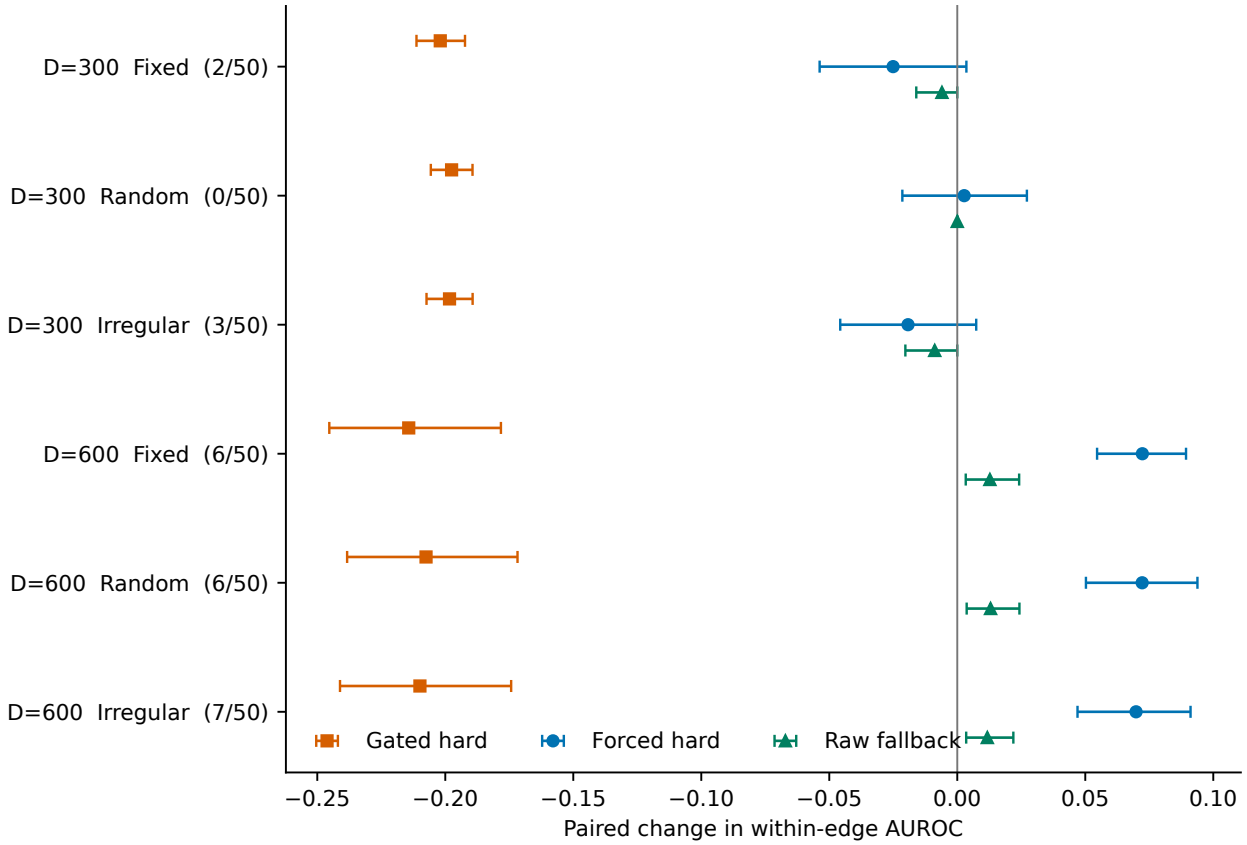


Figure 4: Study B with the gate held fixed: gated hard with static fallback, forced hard emission and raw fallback. Parentheses give accepted/fitted counts. Intervals are pointwise paired-repeat 95% bootstrap intervals; repeated hard-aggregation contrasts reuse the Study B summaries (Supplement Section S5). Raw fallback emits the raw windows on rejection, separating gate selection from information discarded by the rejection output.

5.4 Disentangling cross-edge gains and within-edge information loss

Pooled ranking gains also arise without grouping (Table 4). In Study A, shrinkage at $\lambda = 0.25$ raises pooled AUROC to 0.804 at $D = 300$ and 0.843 at $D = 600$, while preserving the respective raw within-edge temporal AUROCs 0.707 and 0.767. Weights 0.5 and 0.75 also improve pooled ranking. Every positive shrinkage weight preserves within-edge comparisons, including ties, across the A–C predictions. Equation (8) attributes these gains to cross-edge reordering.

Window residuals improve the hard-emission comparison. At $D = 300$, residual weights 0.25, 0.5 and 0.75 yield temporal gains over raw of $+0.010$ $[0.001, 0.020]$, $+0.009$ $[0.003, 0.015]$ and $+0.006$ $[0.003, 0.008]$, respectively, compared with hard emission's -0.069. At $D = 600$, weight 0.25 yields $+0.057$ $[0.050, 0.064]$, compared with hard's $+0.038$. These additional analyses report all fixed weights with pointwise intervals and remain outside the primary confirmatory family.

Changing the rejection output explains much of the gated loss in Study B. At $D = 600$, gains with raw fallback are $+0.013$ $[0.003, 0.024]$, $+0.013$ $[0.004, 0.024]$ and $+0.012$ $[0.003, 0.022]$ for the three schedules, compared with static-fallback gains near -0.21. Both rules accept exactly the same candidates. At $D = 300$, the corresponding raw-fallback means are -0.006, 0 and -0.009. Equation (10) separates losses on rejected repeats from the quality of the accepted candidates. The pair decomposition locates a rank jump at the hard endpoint. At the diagnostic weight 10^{-8}, Study A's Lasso within-edge temporal AUROC rises from 0.639 to 0.707 at $D = 300$ and from 0.805 to 0.830 at $D = 600$. The increases 0.068 and 0.025 arise entirely from resolving same-group positive–negative ties; cross-group credit is unchanged. At weight 0.25 in the longer-duration condition, the same-group component remains $+0.025$ while cross-group reordering contributes -0.006, giving $+0.019$ relative to hard emission. The state assignments are identical throughout this comparison. Supplement Section S5 gives the exact pair accounting and the complete residual grid.

Table 4: Score-emission controls on Study A's Lasso scores. Within-edge denotes within-edge temporal AUROC. All fixed non-endpoint λ values are shown; $\lambda = 1$ duplicates raw. Hard is the residual $\lambda = 0$ endpoint. The paired intervals, OLS checks and full Study B grid appear in Supplement Section S5.

Emission	Pooled AUROC, $D = 300$	Within-edge, 300	Pooled AUROC, 600	Within-edge, 600
Raw	0.727	0.707	0.772	0.767
Forced hard	0.780	0.639	0.839	0.805
Shrink 0.25	0.804	0.707	0.843	0.767
Shrink 0.5	0.781	0.707	0.825	0.767
Shrink 0.75	0.753	0.707	0.799	0.767
Residual 0.25	0.779	0.718	0.835	0.824
Residual 0.5	0.766	0.716	0.819	0.809
Residual 0.75	0.747	0.713	0.797	0.789

5.5 Robustness across generators and window configurations

The effect of aggregation is sensitive to window resolution within the same generator (Figure 5). Study C uses exchangeable non-self support with marginal probability 0.18 in both states, no self edges, identical coefficients on shared support and common lagwise radius scaling. These changes are assessed jointly. Thirty independent repeats are paired across two durations and five window settings at $T = 3600$; all 300 fits succeed.

At $h = 300$, stride 150, forced-hard within-edge temporal AUROC gains are -0.047 at $D = 300$ and $+0.037$ at $D = 600$, matching the direction of Study A's contrast. With $h/s = 200/100$, $300/150$ and $400/200$, the gains are $+0.011$, -0.047, -0.139 at $D = 300$, and $+0.072$, $+0.037$, $+0.045$ at $D = 600$. Holding $h = 300$ and using stride 75 gives -0.028 and $+0.100$; stride 300 gives -0.089 and -0.014. The latter $D = 600$ interval is $[-0.049, 0.022]$, so this grid supports neither a positive gain nor a population sign reversal.

Acceptance still matters when rejection preserves raw scores. At $D = 600, h/s = 300/150$, the raw-fallback gain is -0.007 despite the positive all-candidate gain; at 300/300 it is also -0.007. Supplement Section S4 gives the full window grid and gate denominators, and Section S5 reports its score-emission controls. These robustness comparisons are separate from Study A's primary four-test family.

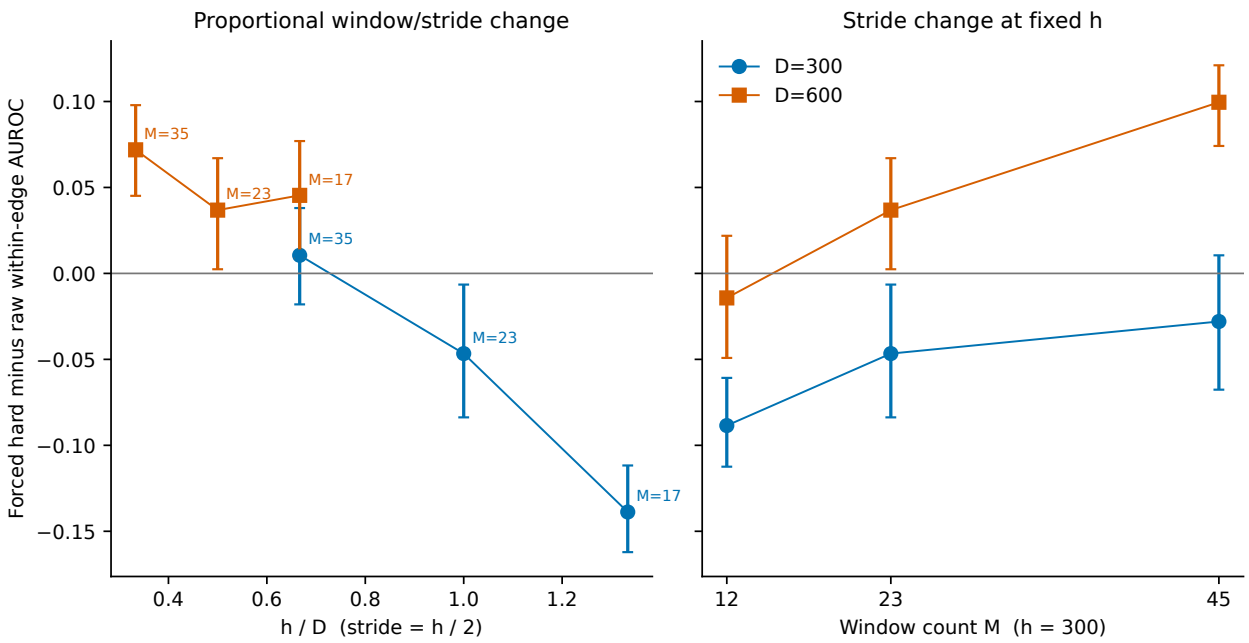


Figure 5: Study C: forced-minus-raw within-edge temporal AUROC in 30 independent repeats per cell. Left varies window size and stride together; labels show window count. Right holds size at 300 and varies stride, giving $M = 12, 23, 45$. Lines connect the tested configurations; intervals are pointwise 95% paired-repeat bootstrap intervals.

5.6 Sensitivity of boundary detection to evaluation conventions

Boundary assessments change with zero-score treatment and threshold reference (Figure 6). For an eligible edge, standard AP exceeds positive-change AP by $\pi_{\text{boundary}}(1-R_+)$, where R_+ is the fraction of true boundary slots with positive predicted strength. At $D = 600$ in Study B, hard-minus-raw standard-AP differences are +0.140, +0.154 and +0.077 for fixed, random and irregular schedules, with pointwise intervals above zero. Positive-change AP gives +0.068, +0.074 and −0.022, with intervals crossing zero.

Amplitude references are also consequential. In B's fixed schedule, q95 F1 favours raw under the common raw-amplitude threshold, with hard-minus-raw -0.077 $[-0.093, -0.060]$, and favours hard under each method's own q95, with $+0.223$ $[0.178, 0.268]$. In C's 300/150 grid at $D = 600$, the respective contrasts are -0.100 $[-0.122, -0.079]$ and $+0.150$ $[0.093, 0.206]$; its standard-AP contrast is $+0.040$ $[-0.020, 0.104]$.

These comparisons describe different operating rules for the same predictions. Method-relative quantiles remove uniform positive rescaling but do not calibrate equal false-alarm rates, and exact zero ties remain relevant. Shrinkage preserves both AP definitions and method-relative F1 on these outputs while potentially lowering common-threshold F1. Supplement Section S5 reports all five fixed quantiles and stable-edge rates, with no selection of an optimal F1 on the evaluation truth.

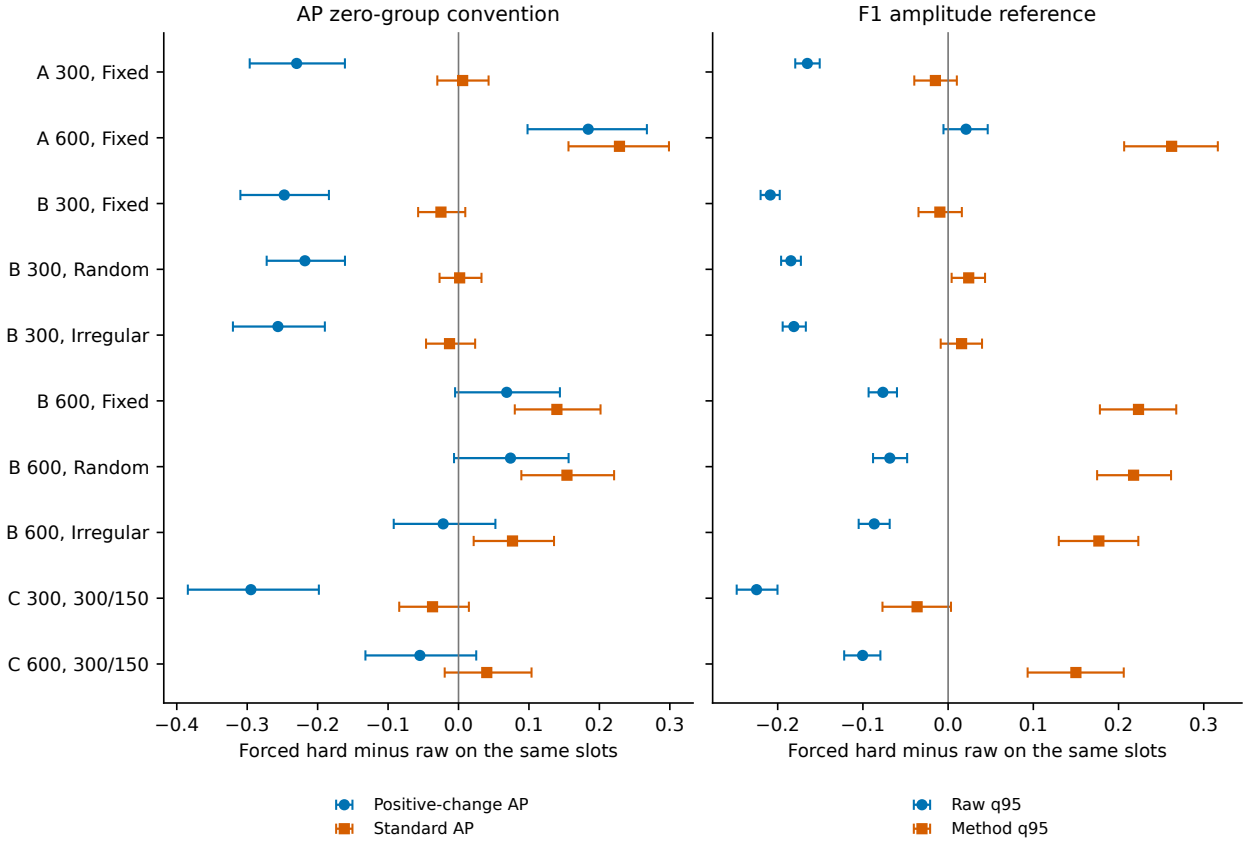


Figure 6: Boundary sensitivity on unchanged predictions, hard minus raw. Left compares positive-change AP with standard AP including the zero-score group. Right compares a shared raw q95 amplitude threshold with each method's own q95. All differences use the same cell's slots; C shows the central 300/150 setting. Intervals are 95% pointwise paired-repeat bootstrap intervals using shared within-cell resamples. Full grids, stable edges and both AP conventions are reported in Supplement Section S5.

5.7 Evaluation beyond recurrent clustering

A segmentation-and-refit framework exhibits a related separation between pooled and temporal gains (Table 5). On Study C's 30 paired repeats at each duration and the 300/150 grid, penalised

PELT partitions the raw window features into contiguous segments. A VAR-Lasso model is then fitted independently on each non-overlapping segment range. This comparison uses our graph-estimation adapter to the external change-point algorithm; it estimates segment count and uses neither recurrent K-means grouping nor the acceptance gate.

Relative to raw windows, PELT plus VAR increases pooled AUROC by $+0.013$ $[0.006, 0.020]$ at $D = 300$ and $+0.008$ $[0.001, 0.014]$ at $D = 600$. Within-edge temporal AUROC differences are $+0.005$ $[-0.020, 0.027]$ and -0.006 $[-0.020, 0.008]$, leaving the temporal comparison inconclusive at both durations. At $D = 600$, positive-change AP gives -0.125 $[-0.205, -0.044]$ while standard AP gives -0.011 $[-0.067, 0.048]$. Thus both the evaluation target and zero-score convention affect the comparison outside recurrent clustering.

All 60 cases and 644 segment refits converge. The fixed configuration yields 9–11 segments on 23 windows, reaching the 11-segment minimum-length cap in 45 cases. Supplement Section S6 provides the settings, counts and paired results, alongside the distinct auxiliary seven-window segmentation study.

Table 5: A different segmentation/refit structure on Study C's 300/150 grid, 30 paired repeats per duration. Within-edge denotes within-edge temporal AUROC. Standard AP includes the zero-score group. F1 uses raw-shared or method-specific q95, without selecting a best operating point. Paired differences use unrounded repeat values and appear in Supplement Section S6.

D	Method	Pooled AUROC	Within-edge	AP_{std}	$F1_{raw}$	$F1_{self}$
300	Raw	0.765	0.707	0.665	0.268	0.268
300	Forced hard	0.872	0.661	0.628	0.043	0.231
300	PELT + VAR	0.779	0.712	0.834	0.321	0.333
600	Raw	0.786	0.761	0.440	0.230	0.230
600	Forced hard	0.901	0.797	0.480	0.130	0.380
600	PELT + VAR	0.794	0.754	0.429	0.222	0.238

6 Discussion and conclusions

Aggregation changes two kinds of comparisons: ordering different edges and ordering the active and inactive windows of one edge. The experiments show why these effects deserve separate evaluation. Hard aggregation improves pooled ranking at both duration configurations in Study A, while its within-edge temporal effect changes sign. Shrinkage achieves pooled gains without grouping, and residual emission restores comparisons that hard maps tie. The fixed local scores make these changes attributable to the emission intervention in the evaluated systems.

The two-state factorisation provides a useful account of relative benefit. The product of state-assignment discrimination and graph-contrast concordance must be interpreted against the temporal ranking already present in raw scores. Better grouping or clearer graph contrast can therefore coexist with little relative gain when the local scores already discriminate active windows well. These ground-truth diagnostics explain an observed configuration; their empirical factors are jointly estimated.

Windowing and execution rules define further conditions of the comparison. Duration changes at fixed record length alter switch coverage, and Study C shows substantial variation across window and stride settings under a common generator. The fixed gate often rejects useful candidates in the longer records. Static fallback then removes temporal variation altogether, whereas raw fallback preserves the local ranking on those rejected cases. Reporting acceptance and the accepted subset's own paired raw comparison makes this execution effect visible.

For evaluation practice, the results support three complementary reports: pooled or target-specific structural ranking, within-edge temporal AUROC, and boundary detection with explicit zero-score and threshold conventions. Stable-edge changes and gate acceptance complete the operational picture. The PELT/refit comparison shows the value of this separation for another partition/refit structure. Reference labels determine which of these reports are possible: the ESS

industrial benchmark supplies a static expert graph, supporting an external-data alignment illustration in Supplement Section S7, while temporal discrimination and boundary recovery require time-indexed truth.

The empirical scope is two-state linear switching systems and local linear scorers. The clustering experiments use a supplied state count, and the PELT adapter estimates a segment count from the same local-score features. Study C's generator modifications are evaluated jointly, so their individual effects remain unresolved. The auxiliary GMM responsibilities are nearly binary and provide limited evidence about diffuse soft aggregation. These conditions identify useful directions for extending the evaluation protocol to other scorers, state structures and nonlinear systems.

Structural ranking, within-edge temporal discrimination and boundary detection answer distinct recovery questions. Shared-score controls and diagnostic accounting make that distinction operational, explaining how an aggregation rule can improve one answer while weakening another.

Data and code availability. Simulation inputs, fitted outputs, analysis code and figure/table generation tools are included in the accompanying reproducibility archive, `information_sciences_reproducibility.zip`. Its README provides separate routes for recomputing saved results and generating and fitting new runs; Supplement Section S8 describes the environment and selected same-seed reproduction checks. The source code and complete archive are available in version `v2026.10.27.1` at https://github.com/xschang1002/latent-state-graph-recovery/releases/tag/v2026.10.27.1.

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Declaration of generative AI use. During the preparation of this work, the author used OpenAI's ChatGPT and Codex to assist with programming, English translation, LaTeX formatting corrections, and data analysis. The author reviewed and verified the AI-assisted outputs as appropriate and takes full responsibility for the accuracy, integrity, and content of the manuscript.

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